with marked anemia, emaciation and loss of weight. Epistaxis (bleeding from nose) and bleeding from gums are common. In Indian patients, the skin on the hands, feet, abdomen, around the mouth and fore – head becomes grayish and dark coloured. This hypo – pigmentation of the skin is unique in Indian patients giving the disease name Kala – azar.

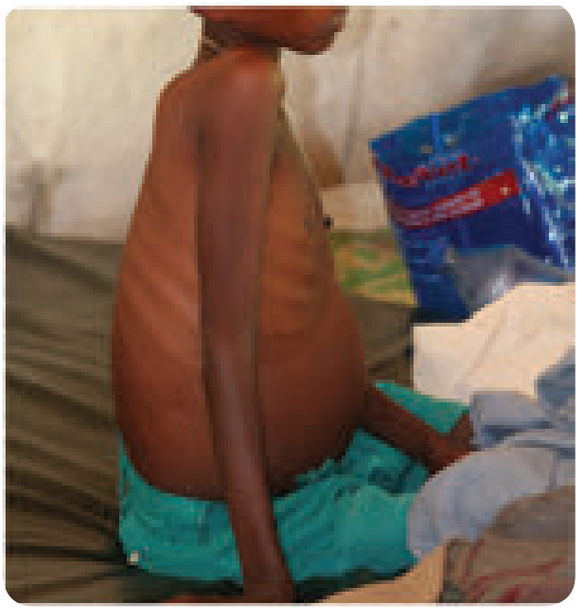


Figure 8.10: Splenomegaly

Post kala – azar dermal leishmaniasis (PKDL): It is a non – ulcerative lesion of the skin, which is seen after completion of treatment of the kala – azar. This condition is characterized by multiple, hypopigmented, erythematous macules involving the face and trunk (Figure 8.11). In Indian forms, PKDL appears after a latent period of 2 years and may even persist as long as 20years, creating a persistent human reservoir of infection.

Laboratory diagnosis

Specimens: Aspiration from spleen, bone marrow, lymph node, liver biopsy and peripheral blood.



Figure 8.11: Post kala – azar dermal leishmaniasis -PKDL

Methods of examination: This includes, microscopy and culture

1. Direct microscopy

The amastigotes of *Leishmania donovani* (known as LD bodies) can be demonstrated in the smears of spleen, bone marrow, liver, lymph node and peripheral blood stained in Leishman, Giemsa or wright stains. Splenic aspiration is the most sensitive method to detect LD bodies. Examination of peripheral blood smear and buffy coat smear is more commonly used to find LD bodies in the circulating monocytes.

2. Culture

Promastigotes are found in the culture media. Tissue samples and aspirates are inoculated in the NNN (Novy–MacNeal–Nicolle) medium for demonstration of promastigotes.

Laboratory diagnosis of kala – azar is briefly discussed in Flowchart 8.5.

Treatment: Pentavalent antimonials are the drugs of choice. Pentamidine, Amphotericin B and Miltefosine (oral drug) are recommended.

TN_GOVT_XII_Micro_Biology_CH08.indd 169 27-02-2019 14:53:05